Huawei Access Node JAY-ACC-68 Major Optical Transport Failure SOP

Node: JAY-ACC-68 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_2889

Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and resolve a Major Optical Transport Failure on Huawei Access Node JAY-ACC-68, located at the Jayanagar Metro Site. The failure domain in question is related to Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) within the Access role. The underlying protocol mechanics at play involve the OCH (Optical Channel) layer of the Optical Transport Network (OTN) protocol stack, which enables the efficient transmission of high-speed optical signals over fiber optic cables. In a Huawei Access Node, the OCH layer is responsible for multiplexing and demultiplexing optical channels, ensuring that data is transmitted correctly between the node and its peers. However, optical transport failures can occur due to various reasons such as SFP/XFP degradation, fiber cuts, or Rx/Tx power loss. These failures can have a significant impact on the performance of the network, leading to packet loss, increased latency, and other service-affecting issues. Therefore, it is essential to diagnose and resolve these failures promptly and efficiently to minimize downtime and ensure high network availability.

Pre-requisite Checks

- Ensure that the Jayanagar Metro Site is accessible to facilitate on-site troubleshooting and maintenance.

- Verify that redundant links are operational to ensure network redundancy and minimize downtime.

- Check the node's power supply and ensure that it is functioning correctly.

Potential Root Causes

Cause	Probability
SFP/XFP Degradation	High
Fiber Cuts	Medium
Rx/Tx Power Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display l1-topology, Expected Output: L1 Topology
Information, Description: This command displays the L1 topology
information of the node, including the optical channels and their
corresponding status. This information can help diagnose optical
transport failures by identifying any issues with the optical
channels.

Command: display l1-channel, Expected Output: L1 Channel Information,
Description: This command displays the L1 channel information of the
```

```
node, including the channel status and the corresponding optical
channel ID. This information can help diagnose optical transport
failures by identifying any issues with the optical channels.

Command: display sfp, Expected Output: SFP Information, Description:
This command displays the SFP information of the node, including the
SFP status and the corresponding optical channel ID. This information
can help diagnose optical transport failures by identifying any issues
with the SFPs.
```

Resolution Steps

Step 1: Step: Replace the faulty SFP/XFP module, Command: replace sfp , Description: Replace the faulty SFP/XFP module with a new one. Ensure that the new module is compatible with the node and the optical channel.

Step 2: Step: Replace the fiber optic cable, Command: replace fiber , Description: Replace the faulty fiber optic cable with a new one. Ensure that the new cable is compatible with the node and the optical channel.

Step 3: Step: Adjust the Rx/Tx power settings, Command: set l1-power , Description: Adjust the Rx/Tx power settings to ensure that the optical channel is operating within the recommended power range.

Post-Check Verification

```
Command: display l1-topology, Expected Output: L1 Topology
Information, Description: Verify that the L1 topology information
displays correctly and that the optical channels are operational.

Command: display l1-channel, Expected Output: L1 Channel Information,
Description: Verify that the L1 channel information displays correctly
and that the channel status is operational.

Command: display sfp, Expected Output: SFP Information, Description:
Verify that the SFP information displays correctly and that the SFP
status is operational.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: undo replace sfp , Description: Restore the original SFP/XFP module if the replacement fails to resolve the issue.

- Step: Restore the original fiber optic cable, Command: undo replace fiber , Description: Restore the original fiber optic cable if the replacement fails to resolve the issue.

- Step: Restore the original Rx/Tx power settings, Command: undo set l1-power , Description: Restore the original Rx/Tx power settings if the adjustment fails to resolve the issue.